

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 5

6. QUANTUM MECHANICS II

• **Tunneling between nanowires / Ex. 6.67**

Two nanowires are separated by $L = 1.3 \text{ nm}$ as measured by STM. Inside the wires the potential energy is zero, but between the wires the potential energy is greater than the electron's energy, $V_0 - E = 0.9 \text{ eV}$. Estimate the probability that the electron passes from one wire to the other.

Solution

For $E < V_0$, the parameter describing the characteristic penetration length is

$$\kappa = \frac{1}{\hbar} \sqrt{2m_e(V_0 - E)} = \frac{2\pi}{hc} \sqrt{2(m_e c^2)(V_0 - E)} = 4.86 \text{ nm}^{-1}$$

which was obtained using $hc = 1240 \text{ eV nm}$ and $m_e c^2 = 511 \text{ keV}$. Since the problem does not give both values of E and V_0 but only their difference, one cannot use the full formula for the transmission

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)} \right]^{-1}$$

but one needs to rely on the approximation

$$T \approx e^{-2\kappa L} = 6.51 \times 10^{-6}.$$

• **STM tunneling / Ex. 6.70**

In Special Topic: Scanning Probe Microscopes, the extreme sensitivity of scanning tunneling microscopes (STM) was described. It was reported that a change in the tunneling gap of only $\Delta L = 0.4 \text{ nm}$ between STM sample and probe can change the tunneling current by a factor of $\xi = 1 \times 10^4$. Check the plausibility of this statement by using the *wide-barrier* approximation in Equation (6.70) to find the difference between the electron energy and barrier height $E - V_0$ that would produce such a situation. Comment on the result.

Solution

When $\kappa L \gg 1$, the transmission can be approximated as

$$T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2\kappa L}$$

The tunneling current is proportional to the transmission coefficient. Suppose there is initially a tunneling current I_0 with a barrier width L_0 . Then the barrier width is increased to $L = L_0 + \Delta L$, resulting in a current decrease to I . The ratio of currents is equal to the ratio of the transmission coefficients as all the factors in the ratio cancel except the exponential, leaving

$$\frac{I}{I_0} = \frac{e^{-2\kappa L}}{e^{-2\kappa L_0}} = e^{-2\kappa \Delta L} = e^{-2 \frac{1}{\hbar} \sqrt{2m_e(V_0 - E)} \Delta L}.$$

In order to solve for the difference $E - V_0$ one takes the natural logarithm, so that

$$\ln \left(\frac{I}{I_0} \right) = -2 \frac{1}{\hbar} \sqrt{2m_e(V_0 - E)} \Delta L$$

and after rearranging

$$V_0 - E = \frac{\hbar^2}{8m_e \Delta L^2} \left[\ln \left(\frac{I}{I_0} \right) \right]^2 = \frac{(hc)^2}{32\pi^2 (m_e c^2) \Delta L^2} \left[\ln \left(\frac{I}{I_0} \right) \right]^2 = 5.0 \text{ eV}.$$

This is entirely reasonable, depending on the materials used for the sample and probe.

• **Scattering and Tunneling**

An electron is accelerated to kinetic energy $E = 5 \text{ eV}$, and scatters on a potential barrier of height V_0 and width $L = 4 \text{ nm}$. Outside the barrier, the potential is zero. Let's denote the regions as I, II, and III.

- Calculate the de Broglie wavelength λ_I of the electron before it hits the barrier. What is the de Broglie wavelength λ_{III} on the other side of the barrier? Calculate the corresponding wavenumbers k_I and k_{III} . Ignore relativistic effects.
- Assuming $V_0 = 4 \text{ eV}$, calculate the de Broglie wavelength λ_{II} , and wavenumber k_{II} inside the barrier.
- Calculate the transmission probability using the equation below. What is the reflection probability?

$$T = \left(1 + \frac{V_0^2 \sin^2(k_{II}L)}{4E(E - V_0)}\right)^{-1}.$$

- V_0 is increased to $V_0 = 6 \text{ eV}$. Verify the system is in the *wide barrier limit*, and calculate the transmission probability using

$$T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0}\right) e^{-2\kappa L}, \quad \kappa = \frac{1}{\hbar} \sqrt{2m_e(V_0 - E)}.$$

Solution

We use the constants

$$hc = 1239.8 \text{ eV nm}, \quad m_e c^2 = 511 \text{ keV}, \quad \lambda = \frac{h}{p}.$$

- The de Broglie wavelengths are calculated using the classical momentum $p = \sqrt{2mK}$. On both sides of the barrier the potential is zero, hence $K = E$ resulting in

$$\lambda_I = \lambda_{III} = \frac{h}{\sqrt{2mK}} = \frac{hc}{\sqrt{2(mc^2)K}} = 0.548 \text{ nm}.$$

The corresponding wavenumber is

$$k_I = k_{III} = \frac{2\pi}{\lambda} = 11.46 \text{ nm}^{-1}.$$

- Inside the barrier $K = E - V_0 = 1 \text{ eV}$. Using this, $\lambda_{II} = 1.226 \text{ nm}$ and $k_{II} = 5.123 \text{ nm}^{-1}$.
- Substituting into the formula, we find $T = 0.557$, which means about 56% chance of the electron going through the barrier.
- Using the formula, we find $\kappa = 5.123 \text{ nm}^{-1}$. The wide barrier limit is applicable if $\kappa L \gg 1$, substituting the numbers we find $\kappa L = 20.49$, which we consider large when appearing in an exponential. Using the transmission formula in the wide barrier limit, we find $T = 3.519 \times 10^{-18}$, which is extremely low.